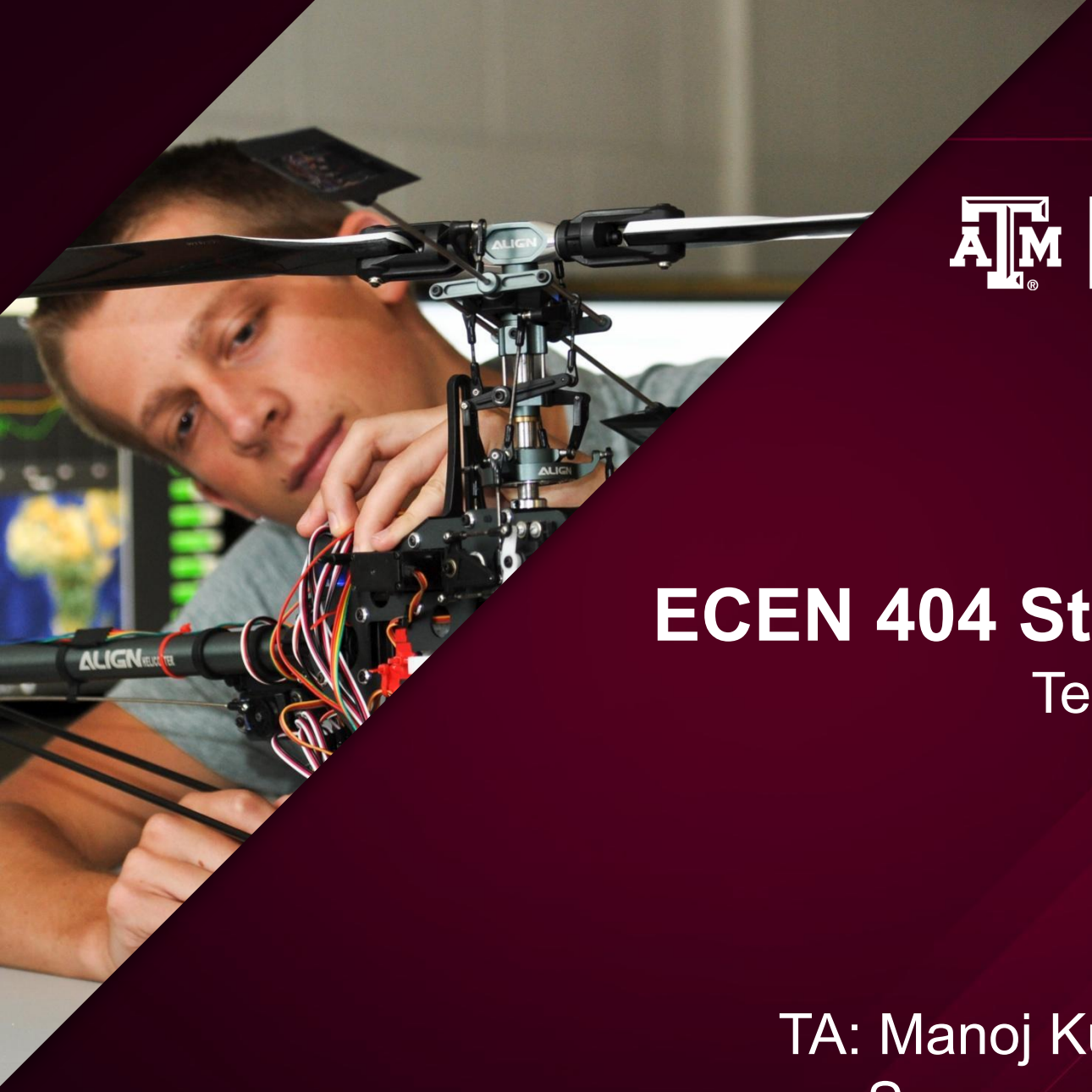

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ECEN 404 Status Update 4

Team 12: TI-Amplifier
Braeden Murdock
Cole Keating
Haotian Guan
Jonathan Roman
TA: Manoj Kumar Balamurugan
Sponsor: Texas Instruments

Project Timeline

Update and order the second iteration of the integrated PCB

Continue debugging power, analog, and digital subsystems
Address integration issues discovered during system testing

Continue Integration

Finish soldering and test functionality of combined board
Completed the integration of app and MCU

02/23/26

03/02/26

03/09/26

03/16/26

Receive updated PCB and continue integration / continue to debug first iteration

Continue debugging UI
Begin soldering the second iteration of the board

Continue Integration

Keep testing the functionality of the combined PCB.

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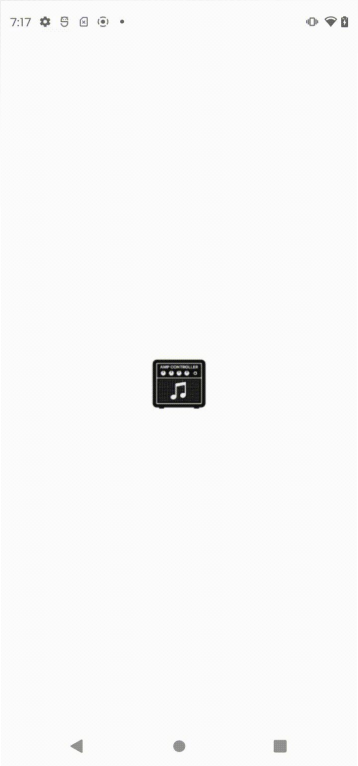
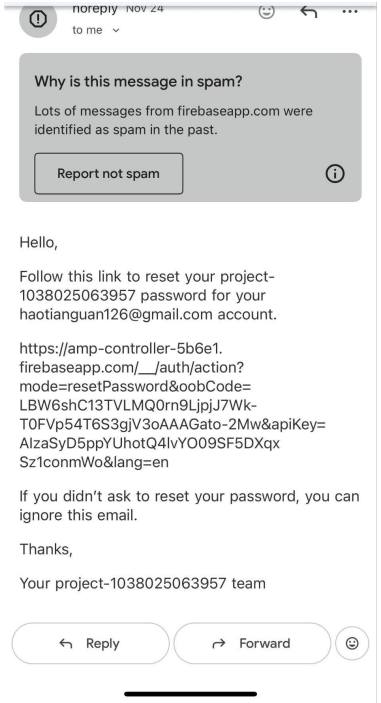
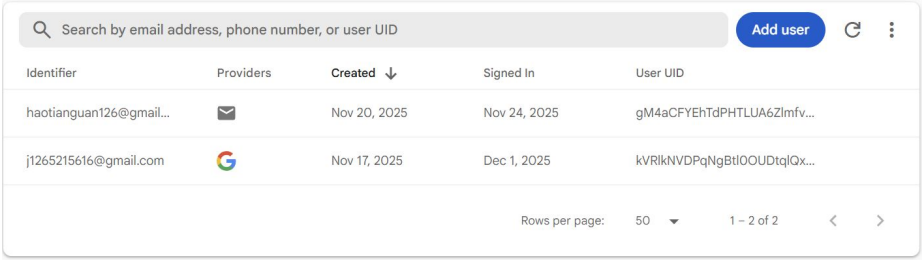
Subsystem 1 - Mobile App

Haotian Guan

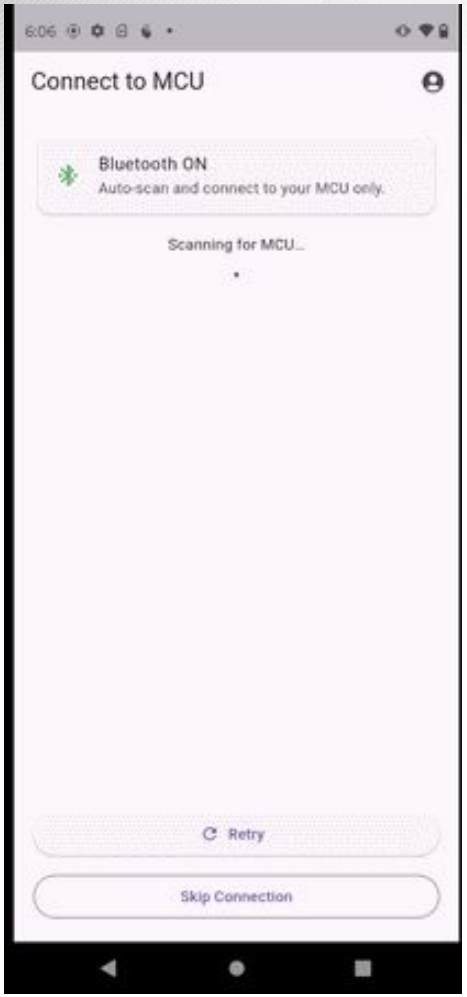






Subsystem 1 - Mobile App

Haotian Guan




```
D/[FBP-Android](17452): [FBP]   status: 0GATT_SUCCESS
D/BluetoothGatt(17452): onConnectionUpdated() - Device=74:4D:BD:68:61:3E interval=39 latency=0 timeout=500 status=0
D/[FBP-Android](17452): [FBP] onMethodCall: setNotifyValue
D/BluetoothGatt(17452): setCharacteristicNotification() - uuid: 00002a05-0000-1000-8000-00805f9b34fb enable: true
D/[FBP-Android](17452): [FBP] onDescriptorWrite:
D/[FBP-Android](17452): [FBP]   chr: 2a05
D/[FBP-Android](17452): [FBP]   desc: 2902
D/[FBP-Android](17452): [FBP]   status: GATT_SUCCESS (0)
D/[FBP-Android](17452): [FBP] onMethodCall: writeCharacteristic
D/[FBP-Android](17452): [FBP] onCharacteristicWrite:
D/[FBP-Android](17452): [FBP]   chr: df170f02-3641-4594-806d-c113a27ce6cb
D/[FBP-Android](17452): [FBP]   status: GATT_SUCCESS (0)
I/flutter (17452): [BLE WRITE] port=2 value=13
I/flutter (17452): [FBP] stopScan: already stopped
D/[FBP-Android](17452): [FBP] onMethodCall: writeCharacteristic
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D/[FBP-Android](17452): [FBP] onCharacteristicWrite:
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D/[FBP-Android](17452): [FBP] onCharacteristicWrite:
D/[FBP-Android](17452): [FBP]   chr: df170f02-3641-4594-806d-c113a27ce6cb
D/[FBP-Android](17452): [FBP]   status: GATT_SUCCESS (0)
I/flutter (17452): [BLE WRITE] port=3 value=0
D/BluetoothGatt(17452): onClientConnectionState() - status=0 clientIf=6 device=74:4D:BD:68:61:3E
D/[FBP-Android](17452): [FBP] onConnectionStateChange:disconnected
D/[FBP-Android](17452): [FBP]   status: SUCCESS
D/BluetoothGatt(17452): close()
```

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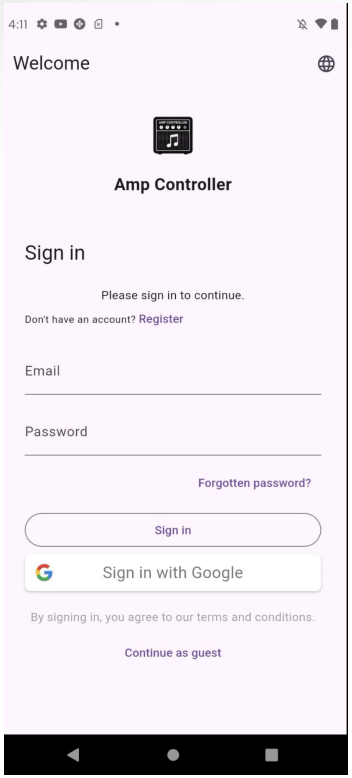
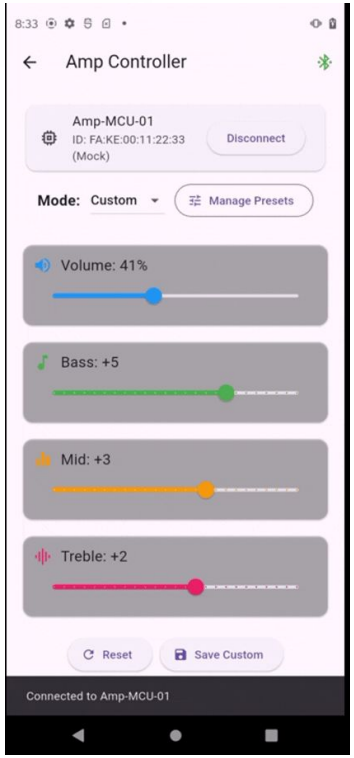
Subsystem 1 - Mobile App

Haotian Guan

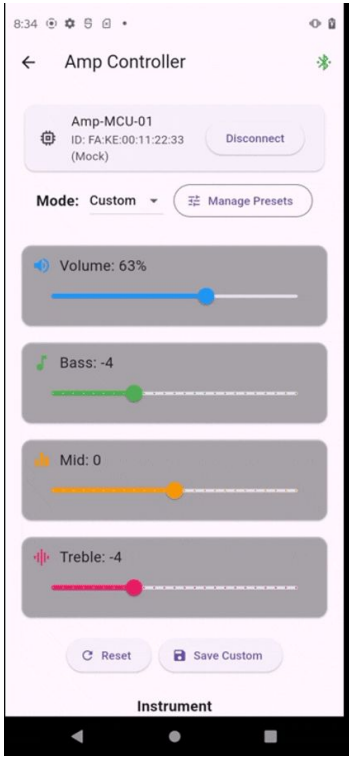
Localization



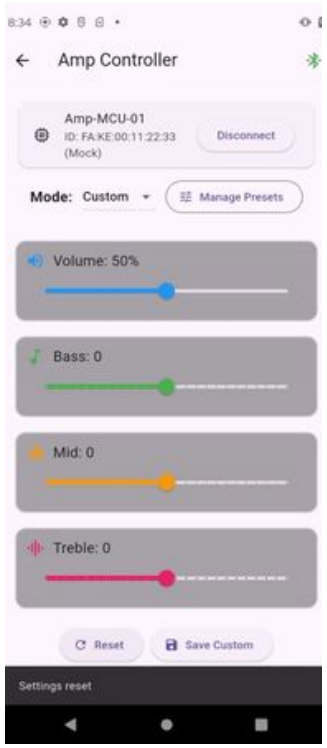
Volume and EQ Control



Preset Management



Electrical Piano



Subsystem 1 - Mobile App

Haotian Guan

Accomplishments since last presentation 14 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Completed the integration between app and MCU • Fixed the issue where returning to the connection page after every time modifying parameters was caused by BLE connection detector	• Continue testing and debugging

Subsystem 2 - Microcontroller

Jonathan Roman

```
bool ar_out[16] = {0,0,0,0,0,0,0,0,
                   0,0,0,0,0,0,0,0};

void ByteArray(uint8_t datain){
  for (int i = 0; i < 16; i++){
    ar_out[i] = false;
  }

  //if (datain >= 128) {ar_out[8] = 1}
  if (datain >= 64) {ar_out[9] = 1;}
  if (datain >= 32) {ar_out[10] = 1;}
  if (datain >= 16) {ar_out[11] = 1;}
  if (datain >= 8) {ar_out[12] = 1;}
  if (datain >= 4) {ar_out[13] = 1;}
  if (datain >= 2) {ar_out[14] = 1;}
  if (datain >= 1) {ar_out[15] = 1;}

}
```

```
/*
Translates the state data based on which POT it should be sent to
If the port is 0 (spi_r[0]) it correlates to volume, on a 100 point scale
Otherwise (the port is 1-3) it correlates to EQ, on a 30 point scale [-15, 15]
*/
uint8_t ByteTranslation (uint8_t input_port, uint8_t input_data) {
  int processed_data = 0;

  if (input_port == 0x00) {
    processed_data = 1.27 * int(input_data);
  } else {
    processed_data = int(input_data) * 4.234;
  }

  return processed_data;

}
```

Subsystem 2 - Microcontroller

Jonathan Roman

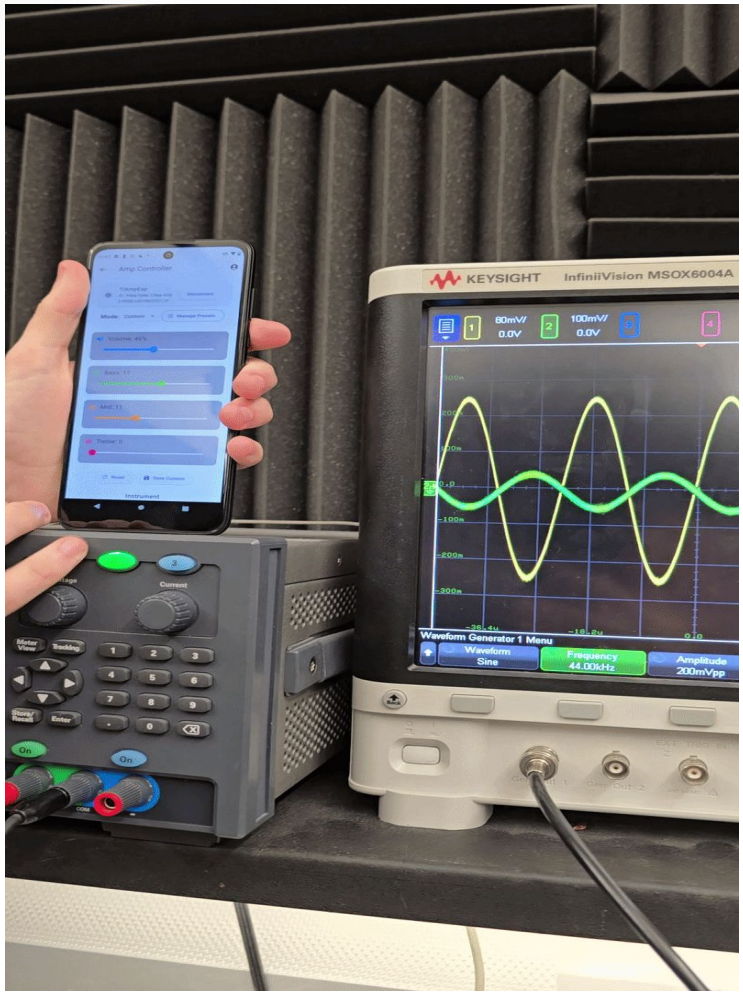
```
//Function used for the actual SPI send
//*spi should be the specific index of spi_r correlating to the correct pot
//data should be processed and on a 256 scale (uint8_t)
void spiCommand(SPIClass *spi, uint8_t data) {
  spi->beginTransaction(SPISettings(spiClk, MSBFIRST, SPI_MODE0)); //configurs parameters to send data

  //sets enable to low (signals beginning of data)
  digitalWrite(spi->pinSS(), LOW);
  spi->transfer(0x00); //*NEW* hopefully this fixes the pot error
  received = spi->transfer(data);        //data transfer
  //sets enable to high (signals end of data)
  digitalWrite(spi->pinSS(), HIGH);

  //ends data transaction
  spi->endTransaction();

}
```

Subsystem 2 - Microcontroller

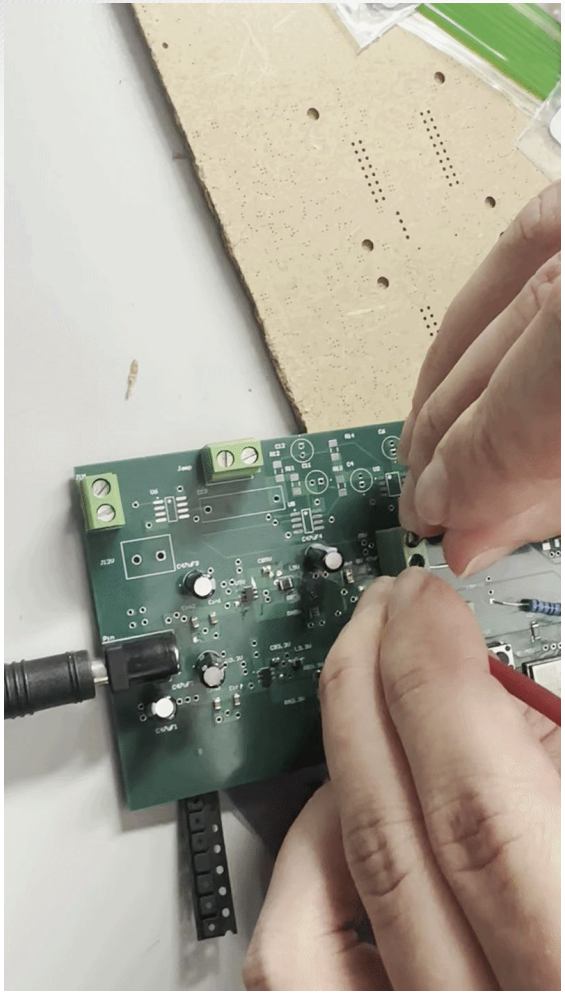
Jonathan Roman

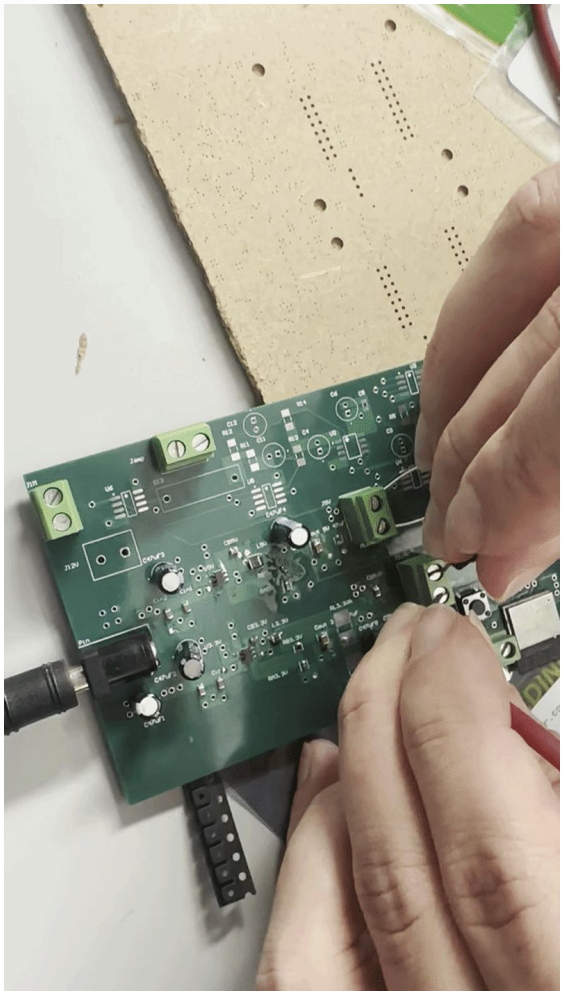
Accomplishments since last presentation 18 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Tested and Refined individual Subsystem Board against Dev Board • Debugged and Reworked code to fix SPI send	• Continue testing on the integrated board • Assemble and Debug Integrated board

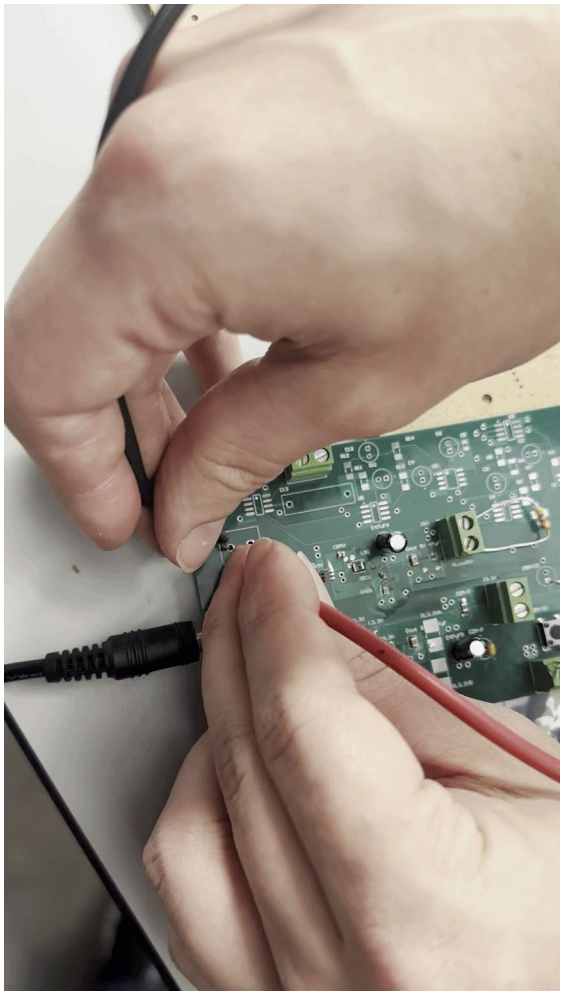
Integrated Board



Integrated Board Subsystem tests







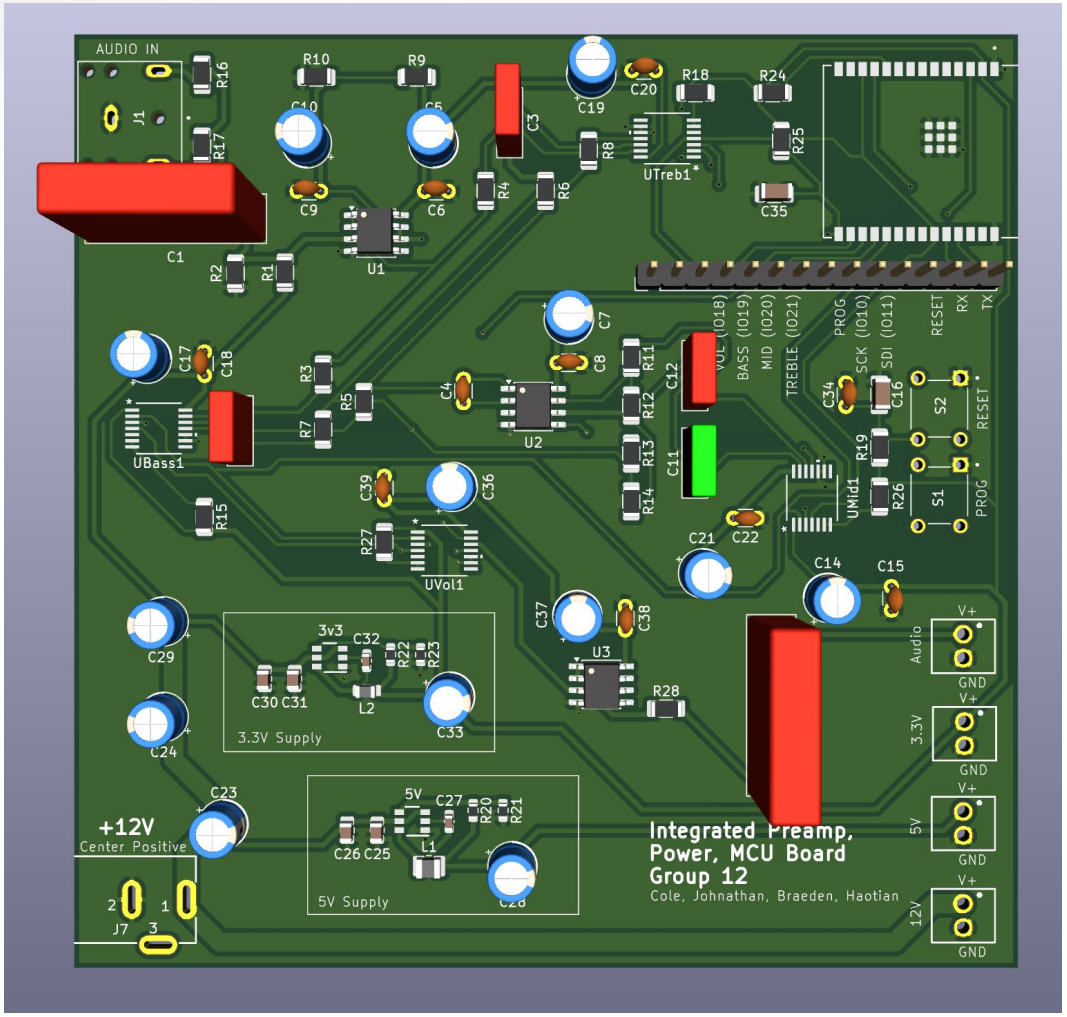
Subsystem 3 - Power System

Braeden Murdock

Accomplishments last presentation 22 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Assisted in finding bug in code that was revealed to be source of failure for MCU sending data to pots • Found reason MCU pin burned out	• Build and get newly acquired pcb working using the knowledge gained from testing current integrated boad

Subsystem 4 - Next Iteration of Integrated PCB

Cole Keating



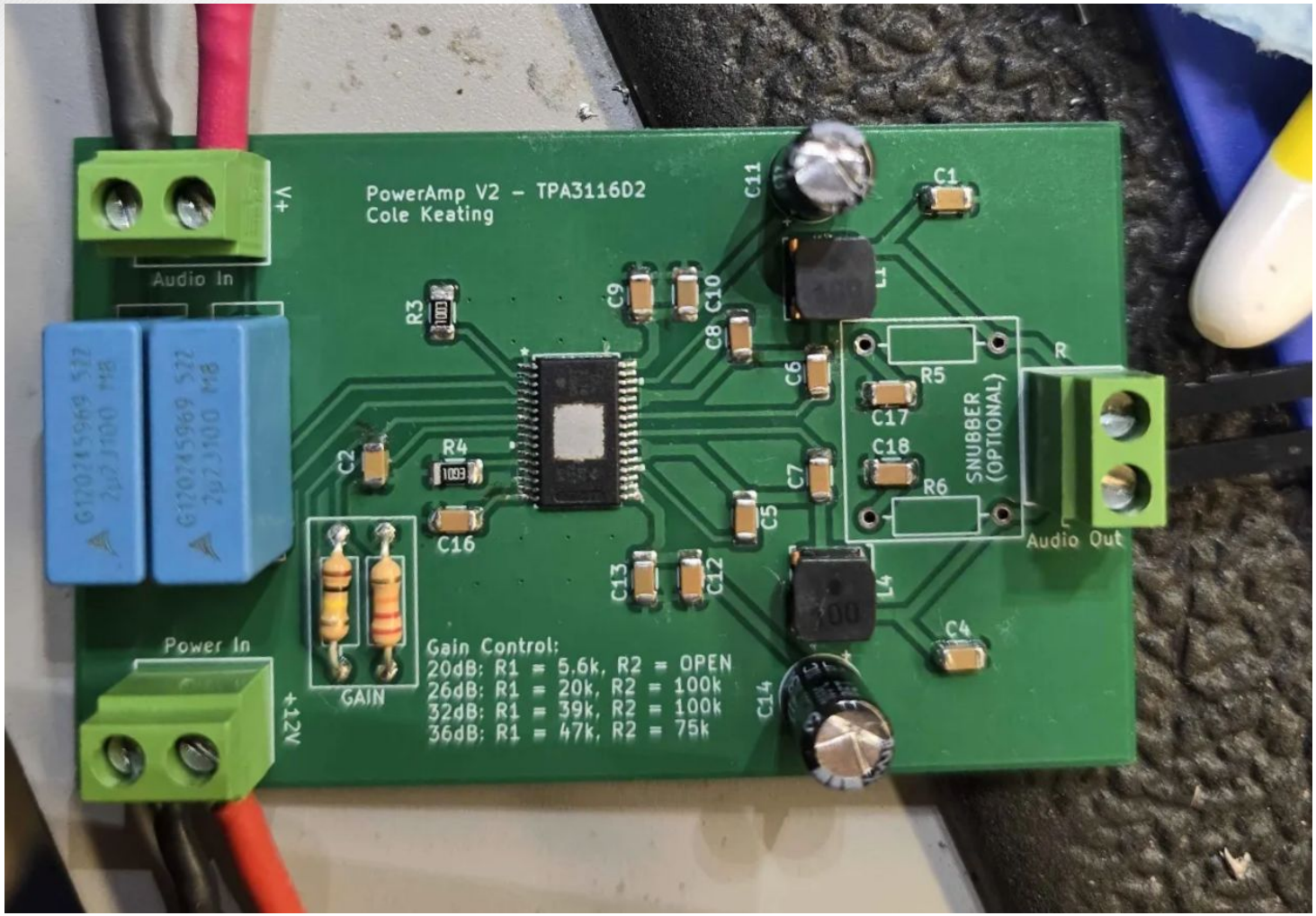
Subsystem 4 - Next Iteration of Integrated PCB

Cole Keating

- Newest revision addressing issues found during testing of original integrated PCB
- New footprint for hand-solderable ESP32, added missing pullup resistors that had to be bodged on original design
- Addresses issues regarding voltage levels for digital pots, board redesigned for higher voltage model of potentiometer
- Added final op amp buffer to address signal degradation issues observed during testing

Subsystem 4 - Assembled Power Amp

Cole Keating



Subsystem 4

Cole Keating

Accomplishments since last presentation 32 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Tested initial integrated design, verified functionality of digital pot control • Designed new integrated PCB with several fixes for issues that arose during testing • Assembled and tested Power Amp, verified output levels and thermal performance	• Current revision of integrated PCB ordered before spring break, should arrive early this week • Power Amp board needs to be assembled - this week

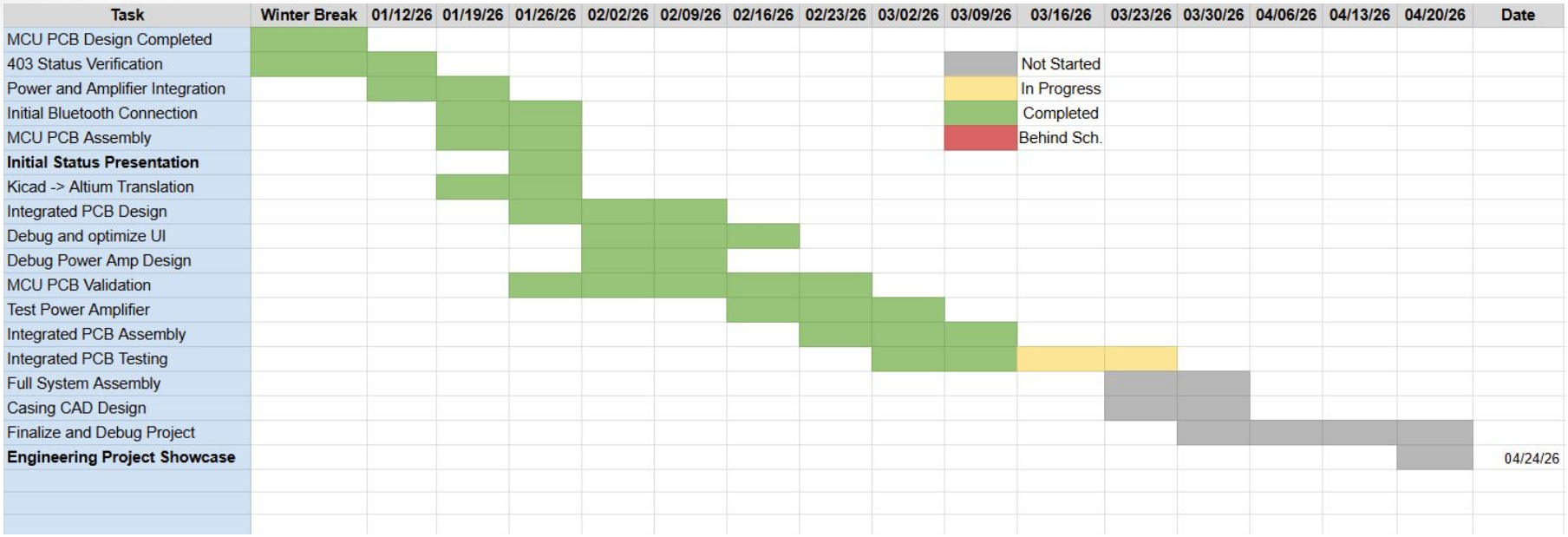
Validation Plan

3	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.2.3.1	Verify input signal levels	Based on research, standard line level signal is 1.4Vpeak, need to ensure this as this is what preamp is biased around	Connect 3.5mm audio cable to source computer, Measure peak voltages to ensure amp isn't being over or underdriven	Tested	Cole Keating
3.2.3.2	Measure power amp heat dissipation	TPA3116 should not enter thermal protection during test	Place cooling solution on chip, power a 8 Ohm load for 30+ minutes without the FAULT pin on the IC going high (indicating thermal protection)	Tested	Cole Keating
3.2.3.3	Verify preamp output signal levels	Preamp output should never exceed 3Vpeak - From MAX9744 datasheet. Signal at max gain should not exceed this	Make sure 0dB bias level is below 10dB of 3V. Set input signal to 1.4V at a mixer frequency. Move potentiometer for max gain, verify output does not exceed 3V peak on oscilloscope.	Tested	Cole Keating
3.2.3.4	Verify power amp output signal level	Power amp output should be able to drive a 8 ohm speaker at 5W without clipping.	Place load on output, verify output levels do not clip.	Tested	Cole Keating
3.2.3.5	Verify THD parameters	THD < 5% at 1WRMS, THD <0.1% at 0.1W RMS	Use FRA mode on oscilloscope while driving a 1W and 0.1W load. Record THD levels.	Tested	Cole Keating
3.2.3.6, 3.2.4.2	Integration of MCU EQ Control	MCU is able to command digital potentiometer. Digital potentiometer controls preamp mixing, volume.	Connect digital potentiometers to pin headers on preamp. Run MCU test code that sweeps potentiometers. Verify digital potentiometer controls preamp, volume	Test in 404	Cole Keating Jonathan Roman
3.2.2	Verify Power levels	Based on research and simulations nothing in the circuit should exceed 2 Watts of power	Connect Power management subsystem to Power input and measure all outputs to make sure they are within the criteria	Tested	Braeden Murdock
3.2.2.1	Verify input voltage levels	Each subsystem needs varous input voltage supplies which are 12V, 5V, and 3.3V. The power management subsystem should output all desired levels of voltage	Measure output voltages of power management subsytem when powered on, by measureing the connectors that would lead to the other subsystems that need the Voltages described previously	Tested	Braeden Murdock
3.2.2.2	Verify input current	Each subsystem has current draw that needs to be matched for the system to work. These range from 16 mA, 0.5 A and 4mA	Set up load resistors that will have the required voltage draw and measure the current through each to make sure that the power management subsystem can match	Tested	Braeden Murdock

Validation Plan Continued

3.2.4.1	Verify Wireless Connection	MCU is able to connect wirelessly to a mobile device. Mobile Device is used to send signals translated by the MCU	Power the MCU with a 3.3V Source. Connect through a generic Bluetooth integration. Send and recieve signals	Tested	Jonathan Roman
3.2.4.1	Verify Wireless Connection With Mobile App	MCU is able to connect wirelessly to the Mobile App subsystem. Mobile App is used to send signals translated by the MCU	Power the MCU with a 3.3V Source. Connect through integrated Mobile App. Recieve and output signals	Tested	Hoation Guan Jonathan Roman
3.2.4.2	Verify Signal Translation	MCU correctly translates input signals to respective SPI output.	Connect wirelessly to the MCU. Send preset signal pairs and verify output	Tested	Jonathan Roman
3.2.4.2	Verify PCB implementation	MCU behaves as expected on PCB implementation	Connect wirelessly to the PCB integrated MCU. Send preset signal pairs and verify output	TESTING/Test in 404	Jonathan Roman
3.2.1.1	Verify the user feedback	Ensure UI give an feedback when use adjust the app setting.	When sliding the control slider, a small prompt appears below indicating the change has been applied. Additionally, a touch effect is displayed when the slider is touched.	Tested	Haotian Guan
3.2.1.2	Verify data saving	Ensure the data stored in the app database, checking database won't lose any data once close the app.	A data storage page has been set up to save all changes, allowing users to view them at any time.	Tested	Haotian Guan
3.2.1.3	Verify the login page	Ensure users can log in using their Google account or register by using the email. Confirm the firebase stored the accounts.	Use Firebase for Flutter to set up a login page with Google accounts. Click “Sign in as guest” to use the app immediately.	Tested	Haotian Guan
3.2.1.4	Verify the BLE connection	Ensure standardized commands sent to the MCU via BLE	Using Flutter's flutter_blue_plus, find the Service UUID and Characteristic UUID of the MCU, and then the App sends 2 bytes [portal, value] to MCU, where portal select valume/bass/mid/treble and value is the user-selected level.	Tested	Haotian Guan Jonathan Roman
3.2.1.5	Verify the BLE detector	Ensure the mobile app UI return to the connection page once disconnection with MCU	Returning back to connection page once the MCU disconnection with mobile app, and exceeding the notification will alert the user.	Tested	Haotian Guan

Execution Plan



Team 12 – TI-Amplifier

Braeden Murdock
Cole Keating
Haotian Guan
Jonathan Roman

Thank you for your attention!
We appreciate your time and interest in our project.
We're happy to take any questions.